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F78AB
Actuarial and Financial Mathematics

Chapter 3: Cashflows and Annuities

LEADERS IN IDEAS AND SOLUTIONS

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3.1 Accumulating and Discounting Cashflows

A series of **certain** cashflows is determined by:

– the times at which they will be paid which we denote t_1, t_2, … t_n ; and

– their amounts.

We denote the amount of the cashflow payable at time t_r by c_r or $c(t_r)$, where r = *1, 2, 3, …* n

Total cashflows = $\sum_{r=1}^{n} c_r$

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3.1 Accumulating and Discounting Cashflows

Principle

- Two or more cashflows paid at different times can be compared or combined *if and only if* we have allowed for the time value of money by finding their accumulations or present values at the same point of time.

Definition

- The value at time t of the series of cashflows defined above is:

$$\sum_{r=1}^{n} c_r V(t, t_r)$$

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3.1 Accumulating and Discounting Cashflows

E.g. 3.1.1

Suppose the effective rate of interest is 4% per time unit, and the cashflows are +1000 at time t_1=1 and +1000 at t_2=3. Calculate:

a) The value of these cashflows at t=0.

b) The value of these cashflows at t=4.

c) The value of these cashflows at t=2.

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3.1 Accumulating and Discounting Cashflows

E.g. 3.1.2 (You try)

Suppose the effective rate of interest is 6% per time unit, and the cashflows are +100 at time t_1=1, -20 at time t_2=2, +50 at time t_3=3 and -10 at t_4=5. Calculate:

a) The accumulation of these cashflows at t=6.

b) The present value of these cashflows at t=0.

c) The value of these cashflows at t=3.

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3.1 Accumulating and Discounting Cashflows

- Once we have calculated the value of a series of cashflows at any one time t, we can find its value at any other time, s, just by applying the appropriate factor $V(s, t)$.
- Given the value of a series of cashflows at time t is,

$$\sum_{r=1}^{n} c_r V(t, t_r)$$

- then its value at time s is,

$$V(s, t) \times (\text{value of cashflows at time } t)$$

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3.1 Accumulating and Discounting Cashflows cont..

- A proof of the above expression appears in Chapter 3 of the notes

- Intuitively you should understand this concept by considering the timeline, the *cashflows, interest rates and time periods over which you are discounting and projecting*

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3.1 Accumulating and Discounting Cashflows

E.g. 3.1.3

Suppose the effective rate of interest is 3% per annum during 2019 and 2020, and 4% per annum thereafter, and that the cashflows are +100 at 1/1/19 and -10 at 1/1/20, 1/1/21, 1/1/22 and 1/1/23, what is the value of these cashflows on

a) 1/1/2024?

b) 1/1/2019?

c) 1/7/2021?

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3.2 Level Annuities-Certain

- A regular series of cashflows is called an **annuity.**
- When these cashflows are certain, we call this an **annuity-certain**.
- If the amount of every cashflow is the same then we call it a **level annuity-certain**.

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3.2 Level Annuities-Certain

- The simplest level annuity has cashflows of 1 at the end of each of the next n time units.
- This is an annuity payable in arrears and is commonly referred to as **level immediate annuity-certain**.
- The number of time units for which the annuity is payable is known as the term of the annuity.

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3.2 Level Annuities-Certain

- Consider a series of n payments, each of amount 1, made at the end of each of the n time periods. Suppose the effective interest rate per unit of time is i.
- The present value at time 0 of this annuity is

$$a_{\overline{n}|}^{i} = \frac{1-v^n}{i}$$

- The accumulation at time n of this annuity is

$$s_{\overline{n}|}^{i} = \frac{(1+i)^n-1}{i}$$

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Proof of first formula:

Observe:

$$(1-v)(1+v+v^2+\cdots+v^{n-1}) = 1-v^n$$

So (for *v* not equal to 1):

$$1+v+v^2+\cdots+v^{n-1} = \frac{1-v^n}{1-v}$$

$$a_{\overline{n}|}^{i} = v+v^2+\cdots+v^n = \frac{v(1-v^n)}{1-v} = \frac{1-v^n}{i}$$

since

$$\frac{1-v}{v} = \frac{1}{v} - 1 = 1+i-1 = i$$

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Example 3.2.1

- Write down the standard symbol for the following and evaluate it at the rate of interest of 12% per annum effective.
  i. The accumulation at time 20 years of £1,000 per annum payable yearly in arrears from now, for twenty years.
  ii. The present value now of £7,000 payable yearly in arrears for the next 15 years.
  iii. The present value now of £500 payable at the end of each of the next 14 half-years.

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3.3 Level Annuities-Due

- An **annuity-due** is an annuity where the payments are made at the start/beginning of each time unit.
- We say that payments are made in advance.
- If n level payments are made then the first payment is made at time 0 and the final payment is at time n-1.

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3.3 Level Annuities-Due

- Consider a series of n payments, each of amount 1, made at the beginning of each of the n time periods. Suppose the effective interest rate per unit of time is i.
- The present value at time 0 of this annuity is

$$\ddot{a}_{\overline{n}|}^{i} = 1 + v + v^2 + \cdots + v^{n-1} = \frac{1-v^n}{1-v} = \frac{1-v^n}{d}$$

- The accumulation at time n of this annuity is

$$\ddot{s}_{\overline{n}|}^{i} = \frac{(1+i)^n - 1}{d}$$

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Example 3.3.1

- Write down the standard (actuarial) symbol for the following and evaluate it at the rate of interest of 12% per annum effective.
  i. The accumulation at time 20 years of £1,000 per annum payable yearly in advance from now, for twenty years.
  ii. The present value now of £7,000 payable yearly in advance for the next 15 years.
  iii. Find the present value of £500 payable at the start of each of the next 14 half-years.

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Quick Question

Which is larger:

$$\ddot{a}_{\overline{n}|} \quad \text{OR} \quad a_{\overline{n}|} \, ?$$

$$\ddot{s}_{\overline{n}|} \quad \text{OR} \quad s_{\overline{n}|} \, ?$$

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3.4 Some Important Relationships

$$\ddot{s}_{\overline{n}|} = (1 + i)\, s_{\overline{n}|}$$

$$\ddot{a}_{\overline{n}|} = (1 + i)\, a_{\overline{n}|}$$

$$s_{\overline{n}|} = 1 + \ddot{s}_{\overline{n-1}|}$$

$$\ddot{a}_{\overline{n}|} = 1 + a_{\overline{n-1}|}$$

You should be able to derive each of these relationships by considering the cashflows and the timing of the calculations along the timeline

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3.5 Perpetuities

- A **perpetuity** is an annuity that never ceases payment; it has an infinite term.
- Consider perpetuities with level payments of 1 per time unit.
- These can be payable at the start/beginning (**perpetuity-due**) or the end of the time unit.
- Their present values have symbols in the international actuarial notation.

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3.5 Perpetuities

- Let the effective rate of interest per time unit be i (> 0), a constant. Then
  - the present value at time 0 of a perpetuity of 1 per time unit payable in **arrears** is

$$a_{\overline{\infty}|}^{i} = \lim_{n\to\infty} a_{\overline{n}|}^{i} = \lim_{n\to\infty} \frac{1-v^n}{i} = \frac{1}{i}$$

  - the present value at time 0 of a perpetuity of 1 per time unit payable in **advance** is

$$\ddot{a}_{\overline{\infty}|}^{i} = \frac{1}{d}$$

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Example 3.5.1

- Let the effective rate of interest be 5% per time unit.

Find $a_{\overline{\infty}|}$

Solution:

Find $\ddot{a}_{\overline{\infty}|}$

Solution:

Example 3.5.1

- Let the effective rate of interest be 5% per time unit.
  - Find the value at time t = 10 of a perpetuity of £100 per time unit payable annually in arrears and commencing now.

Solution: Value at time t = 10 is

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3.6 Deferred Annuities

- Sometimes an annuity does not start immediately but is deferred (postponed) for a period of time.
  - **An employer might promise to pay an employee a pension when he retires.**
- The present value at time 0 of an annuity of 1 per time unit payable in arrears for n time units, commencing after m time units, where the effective rate of interest is i per time unit is

$$ {}_{m|}a^{i}_{\overline{n}|} $$

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3.6 Deferred Annuities

- Another way of describing a deferred annuity:
  - The value at time 0 of a series of n payments, each of amount 1, made at times $(m+1)$, $(m+2)$, $(m+3)$, . . . $(m+n)$.
- The present value of this deferred annuity is

$$ {}_{m|}a^{i}_{\overline{n}|} = v^{m} a^{i}_{\overline{n}|} = a^{i}_{\overline{m+n}|} - a^{i}_{\overline{n}|} $$

- If payments are made in advance then

$$ {}_{m|}\ddot{a}^{i}_{\overline{n}|} = v^{m} \ddot{a}^{i}_{\overline{n}|} = \ddot{a}^{i}_{\overline{m+n}|} - \ddot{a}^{i}_{\overline{n}|} $$

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Example 3.6.1

Find the present value at outset of a level annuity of £100 per year, payable annually in arrears for ten years, deferred for ten years, if the effective rate of interest is 10% per annum.

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Example 3.6.2

Find the present value at outset of a level annuity of £100 per year, payable annually in arrears for 10 years, deferred for 10 years, if the effective rate of interest is 10% per annum during the first 10 years and 8% per annum during the last 10 years.

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3.7 Increasing Annuities

- It is sometimes necessary to work with annuities that increase at a regular rate.
- The increase can be Simple (Arithmetic) or Compound (Geometric).

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3.7.1 Simple (Arithmetically) Increasing Annuities

- The simplest form of an increasing annuity is one that increases by the same amount each time unit.
- The simplest such annuity is that which has first payment 1 and then increases by one per time step.
- Consider the timeline for these payments

Increasing annuities - Notation

Let i be the effective rate of interest, assumed constant. Then the following symbols are standard in actuarial practice.

Symbol	Meaning
$(Ia)^{i}_{\overline{n}\rceil}$	The present value at time 0 of the arithmetically increasing annuity paid in arrears.
$(Is)^{i}_{\overline{n}\rceil}$	The accumulation at time n of the arithmetically increasing annuity paid in arrears.
$(I\ddot{a})^{i}_{\overline{n}\rceil}$	The present value at time 0 of the arithmetically increasing annuity paid in advance.
$(I\ddot{s})^{i}_{\overline{n}\rceil}$	The accumulation at time n of the arithmetically increasing annuity paid in advance.

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Important Results

1. $(Ia)^{i}_{\overline{n}|} = \frac{\ddot{a}^{i}_{\overline{n}|} - nv^{n}}{i}$

2. $(Is)^{i}_{\overline{n}|} = \frac{\ddot{s}^{i}_{\overline{n}|} - n}{i}$

3. $(I\ddot{a})^{i}_{\overline{n}|} = \frac{\ddot{a}^{i}_{\overline{n}|} - nv^{n}}{d}$

4. $(I\ddot{s})^{i}_{\overline{n}|} = \frac{\ddot{s}^{i}_{\overline{n}|} - n}{d}$

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Proof of First Result

$$(Ia)^{i}_{\overline{n}|} = v + 2v^2 + \cdots + nv^n$$

$$(1+i)(Ia)^{i}_{\overline{n}|} = 1 + 2v + 3v^2 + \cdots + nv^{n-1}$$

Taking the first equation from the second gives:

$$i(Ia)^{i}_{\overline{n}|} = 1 + v + v^2 + \cdots + v^{n-1} - nv^n$$

and therefore

$$(Ia)^{i}_{\overline{n}|} = \frac{\ddot{a}^{i}_{\overline{n}|} - nv^n}{i}$$

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Example 3.7.1

Let the effective rate of interest be a constant 5% per annum. Find the present value at outset of an annuity payable annually in advance, for ten years, whose first payment is £1 and whose payments increase by £1 per annum.

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Example 3.7.2

Write down, in terms of the standard annuity symbols defined in this chapter, an expression for the present value of an annuity payable annually in arrears for 20 years, whose first payment is £10,000 and whose payments increase by £1,000 each year.

Calculate this present value at an effective rate of interest of 5% per annum.

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Example 3.7.3

Write down, in terms of the standard annuity symbols defined in this chapter, an expression for the present value of an annuity payable annually in advance for 15 years, whose first payment is £30,000 and whose payments decrease by £2,000 each year.

Calculate this present value at an effective rate of interest of 6% per annum.

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3.7.2 Compound (Geometrically) Increasing Annuities

- An annuity which increases by a fixed multiple or factor each time unit.
  - A pension which increases by 5% each year to allow for inflation.
- Such annuities are not included in the international actuarial notation, so they have to be described in words.
- No standard formulae exist for dealing with these, so expressions for values have to be written down from first principles.

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Example 3.7.4

Suppose a person now age 65 has a pension payable annually in arrears, whose first payment is £1 which will be paid on their 66th birthday.
The last payment will be made on their 100th birthday.
To combat inflation, payments will grow by 5% each year; that is, each payment will be 5% greater than the previous payment.
If the effective rate of interest is a constant 7% per annum throughout, what is the present value of this pension now?

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Next chapter…

Chapter 4: Nominal Interest Rates

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Attendance matters!

